AMCC 5160 · LECTURE 07 · 13 OCTOBER 2026

# What's New in *AI Motion*

Music, minutes, and machines · Fall 2026

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# Four stories from this year

01

## The 5-second clip is over

30-second single generations; a real-time minute

Mar – Sep 2026

02

## Music becomes the choreographer

A song and a photo give you a minute of dance

Jul 13, 2026

03

## Motion leaves the screen

A humanoid learns a K-pop dance from phone video

Jun 10, 2026

04

## Sora is gone

What a platform shutdown means for artists

Apr 26 & Sep 24, 2026

# How many shots is this?

Watch the clip. Count the cuts. Then guess how many times the model was run.

▶ Play: Seedance 2.5 showcase (official page, hero video)

seed.bytedance.com/en/seedance2_5

# How top courses treat what's new

| Course | What it does with new work | Tonight |
|---|---|---|
| **Stanford CS348K**<br>Fatahalian · Spring 2026 | A full lecture on how distillation and caching made autoregressive video real-time; guests from WorldLabs, Roblox, NVIDIA | **Story 1** |
| **CMU 16-726**<br>Jun-Yan Zhu · Spring 2025 | Put a same-year fast video paper on the reading list; ends with forensics and data attribution | **Story 2** |
| **MIT 4.562**<br>Nagakura · Fall 2025 | Listed Sora as the video tool in the syllabus | **Story 4** |
| **USC IML-460**<br>Holly Willis · Spring 2026 | Pairs every tool with the controversy around it | **Each “For the room”** |

STORY 1 OF 4

*01*

# The 5-second clip is over

From fragments to scenes, and from rendering to real time

# The unit of AI video is now the scene

~5 s

A typical clip, 2024

→

~15 s

Seedance 2.0, early 2026

→

30 s

Seedance 2.5 · Wan 3.0 · Kling 4.0, one pass

→

~60 s

Helios, generated faster than real time

Durations are per single generation, as stated by vendors or reviewers.

# Who shipped what

| Model | What's new | When |
|---|---|---|
| **Seedance 2.5 · ByteDance** | Built for 30-second storytelling; camera moves and performance blocking | Jul 31, 2026 |
| **Wan 3.0 · Alibaba** | 30-second clips; beta only, no open weights | Public beta, Aug 2026 |
| **Kling 4.0 · Kuaishou** | Announced 3–30 s, up to 4K; Flash tier 20 s at 720p | Flash Sep 28; full model October |
| **Helios · Peking Univ. + ByteDance** | 14B model; ~60 s at 19.5 FPS on one H100; open weights | Mar 2026 |

Seedance 2.5 official page ↗

seed.bytedance.com/en/seedance2_5

# Helios: a minute of video, in real time




Frame from the official Helios feature video

- 14B autoregressive model: writes video forward, chunk by chunk
- ~60 s (1,440 frames) per generation
- 19.5 fps on one H100: close to 24 fps playback
- Open weights

**▶ Play: Helios feature video (YouTube)**

www.youtube.com/watch?v=vd_AgHtOUFQ

**▶ Minute-long demos (project page)**

pku-yuangroup.github.io/Helios-Page

**Try it: Hugging Face demo ↗**

huggingface.co/spaces/BestWishYsh/Helios-14B-RealTime

# Why long video falls apart: drift

## Second 1

The character looks right

→

## Second 20

Small errors feed the next chunk

→

## Second 60

Face, clothes and room have changed

Helios's fix: train on deliberately drifted video, so the model learns to recover instead of patching drift at playback time.

FOR THE ROOM

# If one generation is a whole 30-second scene, what is the director's job now?

And what did the editor just lose?

STORY 2 OF 4

# 02

# Music becomes the choreographer

A song, one photo, a minute of dance

# Wan-Dancer: song in, dance out

A music track

One photo of the dancer

A style word

→

A 1-minute+ dance video, 720p / 30 fps, on the beat

- Released Jul 13, 2026 by Tongyi Lab (Alibaba) as open weights, Apache 2.0
- Five styles: Chinese classical, K-pop, street, tap, Latin

▶ Play: Wan-Dancer demos (project page)

humanaigc.github.io/wan-dancer-project

Paper (arXiv 2607.09581) ↗

arxiv.org/abs/2607.09581

# One dancer, many looks; one song, many dancers




Reference images from the Wan-Dancer project page

- Swap the photo, keep the song: the same dance, a different dancer
- Swap the song, keep the photo: the same dancer, a new dance
- Earlier music-to-dance models fell apart after about 20 seconds

▶ Play: compare with earlier models (§6)

humanaigc.github.io/wan-dancer-project

# The trick: plan the song, then the steps

**Global level: the whole song**

| Intro | Verse | Build | Chorus | Outro |
|---|---|---|---|---|

↓ plans

**Local level: the moves**

Renders each passage's steps, beat by beat

*This is how a choreographer works: structure first, steps second.*

# Pitch a one-take

5 MINUTES · IN PAIRS

1

One song: name it, and say why

2

One photo: who or what dances?

3

One sentence: what happens in the 30 seconds?

Three pairs read theirs aloud. Keep your pitch: it could become Assignment 2.

FOR THE ROOM

# Who is the choreographer here?

The musician? The model? The thousands of dancers in its training videos? The person who picked the song?

STORY 3 OF 4

*03*

# Motion leaves the screen

A phone video becomes a robot's dance

# A humanoid learns a K-pop challenge from phone video




ROBOTIS AI Sapiens dancing the CORTIS “REDRED” challenge (ROBOTIS video thumbnail)

- ROBOTIS (Korea), open-source humanoid AI Sapiens, June 2026
- 1.3 m tall, 34 kg, 23 joints
- Learned from ordinary 2D video: no motion-capture studio

**▶ Play: ROBOTIS dance video (YouTube)**

www.youtube.com/watch?v=NeL2BmgcFJs

**News story (Interesting Engineering) ↗**

interestingengineering.com/ai-robotics/korean-humanoid-r…

# Four steps from phone to robot

1

**Film**

A dancer, on a phone

→

2

**Lift**

Recover 3D joint motion from the 2D video

→

3

**Practise**

Reinforcement learning in simulation fixes balance and timing

→

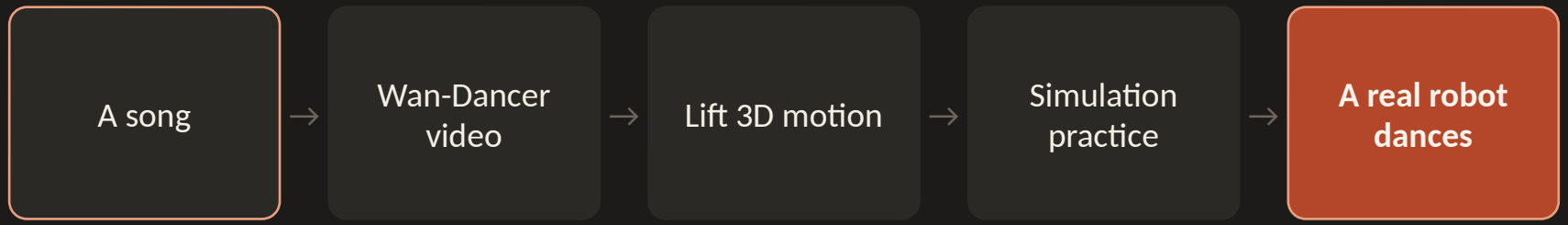
4

**Transfer**

Move the learned skill to the real robot

*Every step is an idea from this course: pose from video, simulation, and a model of the world.*

# A thought experiment




Speculation: as far as public reports go, nobody has shown this end to end. But every piece now exists.

FOR THE ROOM

# If a generated video can teach a real body to move, is that video still “fake”?

What is a dance video for, once machines learn from it?

STORY 4 OF 4

# 04

# Sora is gone

When your tool disappears

# Eighteen months as a product

**Dec 2024**
Sora launches as a product

**Sep 30, 2025**
Sora 2 + a social video app

**Mar 24, 2026**
OpenAI announces the shutdown

**Apr 26, 2026**
App and website close

**Sep 24, 2026**
API switched off, no replacement named

**OpenAI: What to know about the Sora discontinuation ↗**

help.openai.com/en/articles/20001152-what-to-know-about-...

# Last fall, a top syllabus was built on Sora

MIT's Fall 2025 motion-graphics course listed Sora as its video tool. Some of last year's student projects depend on a tool that no longer exists.

# Keep your sources, not just your outputs

- [x] Prompts, word for word
- [x] Seeds and settings
- [x] Reference images, clips and songs (with credit)
- [x] Raw generations, before editing
- [x] Edit project files
- [x] Model name and version, and the date

**Why open weights matter**

Open-weight models (Wan 2.2, Helios, Wan-Dancer) can still be run years from now. Closed apps can't.

FOR THE ROOM

# What does an AI artwork need in order to outlive the tool that made it?

Think of a film, a painting, a game. What survives, and why?

DEBATE · 10 MINUTES

# *“A generated dance that a robot can learn is a real dance.”*

Left side of the room argues for; right side against

Two speakers per side, one minute each

Then the whole room votes by show of hands

# This week

READING 1 · TECHNIQUE

**Helios: real-time long video generation**

READING 2 · ART + TECH

**Wan-Dancer project page**

READING 3 · CULTURE

**What happened to Sora?**

**Quiz 07 · two technique, two art, one reflection · hyang.org/amcc5160/quiz-07**

Assignment 2 is due Oct 31 · Next week, Lecture 08: native audio-video, performance and real-time characters

# All demo links in one place

**STORY 1**

- ▶ Seedance 2.5 official page
- ▶ Helios feature video (YouTube)
- ▶ Helios project page

**STORY 2**

- ▶ Wan-Dancer project page
- ▶ Wan-Dancer weights (Hugging Face)

**STORY 3**

- ▶ ROBOTIS: Learning from Videos (YouTube)
- ▶ AI Sapiens documentation

**STORY 4**

- ▶ OpenAI: Sora discontinuation

# *Music, minutes, and machines*

Motion got longer, found a new choreographer, left the screen, and lost a tool. Keep your sources.

Sources: Seedance 2.5 (ByteDance Seed); Helios (Peking University, ByteDance, arXiv 2603.04379); Wan-Dancer (Tongyi Lab, arXiv 2607.09581); ROBOTIS AI Sapiens; OpenAI Help Center; reviews by Pinggy, Atlas Cloud and Unifically for Wan 3.0 and Kling 4.0 specs. Images: Helios feature video, Wan-Dancer project page, ROBOTIS video thumbnail, used for teaching.